

Sexual selection as a neglected driver of reproductive isolation across the flowering plants

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Abstract

Here, I develop a model for how post-pollination prezygotic reproductive isolation can arise in flowering plants. The mechanism is the number of effective pollen grains competing per ovule in an ovary. When that ratio is high, pollen grains compete for limited fertilisation opportunities. Selection can then favour pollen traits that win fertilisations and maternal traits that filter pollen more strictly. Reproductive isolation arises as a by-product when the evolved pollen and pistil traits of one lineage no longer work well with those of another. The main empirical prediction is directional. Reciprocal crosses should fail more strongly when the high-competition lineage is the maternal parent. I then audit the current data structure and show that the strongest immediate test is an asymmetric reciprocal-cross test, analysed with family-level partial pooling and sensitivity analyses for ovule-number confidence, genetic distance, mating system, measurement type, and phylogenetic non-independence. The current data support a useful first analysis, but not a fully saturated model.

Conclusion

A note on the coloured boxes. *Rule used* states the mathematical rule a step relies on. *Why this step* explains the reasoning behind the move, because the difficulty is rarely the algebra; it is usually seeing why a step is allowed or why we would want it. *Conclusion* states what we have just learned. *Caution* flags a place to be careful. *Definition* and *Working metaphor* do what their names say.

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1 Purpose of this note

I want a model that does three jobs. First, it must state the biological mechanism in a testable form. Second, it must show how ovule number, pollen load, mating system, and pollen-pistil mismatch enter the same set of equations. Third, it must lead to predictions that can be separated from older explanations, especially unilateral incompatibility.

The starting idea is simple. An ovary has a finite number of ovules. A stigma receives pollen. If the number of effective pollen grains is smaller than or similar to the number of ovules, most compatible pollen grains can fertilise, and competition among pollen grains is weak. If the number of effective pollen grains is much larger than the number of ovules, some pollen grains must lose. That losing margin creates selection.

Working metaphor

Think of ovules as seats in a small room. Pollen grains are applicants for those seats. When there are as many applicants as seats, the filter can be loose. When there are many more applicants than seats, the filter can be strict, and even small differences among applicants matter.

The model is not that few ovules directly cause reproductive isolation. The model is that few ovules can move a lineage into a high-competition reproductive state. In that state, selection on pollen performance and maternal filtering becomes stronger. Reproductive isolation can then appear as a passenger of that selection. Ovule number can be thought of as a modulator of the strength of sexual selection in plants.

Conclusion

The causal claim is not

few ovules $\Rightarrow$ reproductive isolation.

The causal claim is

few ovules $\Rightarrow$ higher pollen competition

$\Rightarrow$ stronger pollen and pistil selection

$\Rightarrow$ directional prezygotic isolation.

The middle terms are the mechanism.

Note that this model does not predict the evolution of directional postzygotic isolation, a consideration that we explain below.

2 Biological background

Speciation studies often treat reproductive isolation as something that increases with time. This view works well for many postzygotic barriers. Hybrid sterility and hybrid inviability can increase as substitutions accumulate between lineages. In plants, however, post-pollination prezygotic barriers are not only historical residues. They also depend on the current reproductive arena.

After pollen lands on a stigma, it must hydrate, germinate, enter maternal tissues, grow through the style, respond to guidance cues, reach ovules, and release sperm cells. These steps occur inside maternal tissue. They allow selection on pollen traits and on maternal filtering traits (pers. comm. Marcy Uyenoyama, 2007). The pollen tube pathway in angiosperms has therefore been viewed as a setting for pollen competition and selection (Lora et al., 2016; Lankinen and Karlsson Green, 2015). Reviews of post-pollination barriers also emphasise that pollen-pistil interactions can reject heterospecific pollen and contribute to prezygotic reproductive isolation (Wang and Filatov, 2023).

The important point for this model is that the pistil is not a pipe. It is a selective environment. That environment may become more selective when pollen abundance exceeds ovule abundance.

Conclusion

The pistil is part of the selective environment. This matters because a maternal trait can create a directional barrier. Genetic distance is symmetric. A cross direction is not.

3 Notation

Symbol	Meaning
O_i	Number of ovules per ovary in lineage i .
A_i	Pollen production per relevant unit.
ψ_i	Effective deposition fraction; combines transfer, germination, and timing.
$\delta_i, \gamma_i, \tau_i$	The three parts of ψ_i : transfer to a stigma, germination/entry, temporal overlap.
ι_i	Inbreeding depression in lineage i (mating-system filter model).
$P_{\text{eff},i}$	Effective pollen load: pollen that can compete for the same ovules.
K_i	Effective pollen grains per ovule, $K_i = P_{\text{eff},i}/O_i$.
F_i	Strictness of the maternal filter in lineage i .
D_{ij}	Pollen-pistil mismatch between i and j ; δ_D scales it to genetic distance.
G_{ij}	Genetic distance or divergence between i and j .
C_{ii}	Conspecific reproductive success for maternal lineage i .
$H_{i \leftarrow j}$	Heterospecific success for cross mother i , father j .
$R_{i \leftarrow j}$	Directional RI index for cross mother i , father j .
A_{ij}	Reciprocal asymmetry, $A_{ij} = R_{i \leftarrow j} - R_{j \leftarrow i}$.
t_i	Outcrossing rate of lineage i .
s_i	Self-incompatibility status, $s_i = 1$ for SI, $s_i = 0$ for SC.

Caution

One symbol, one meaning. In particular δ_i is the pollen-deposition fraction, while inbreeding depression is written ι_i . These were the same letter in an earlier draft; they are now distinct.

4 The competition ratio

4.1 Definition

The central quantity is

$$K_i = \frac{P_{\text{eff},i}}{O_i}. \quad (1)$$

$P_{\text{eff},i}$ is not total pollen production, nor even all pollen on the stigma. It is the pollen that can plausibly compete for the same ovules: arriving in the right window, viable, germinated, in the competitive pathway.

Definition

K_i is the number of effective pollen grains per ovule. It is a competition index, not a pollen count.

A useful decomposition is $P_{\text{eff},i} = \psi_i A_i$, with

$$\psi_i = \delta_i \gamma_i \tau_i, \quad (2)$$

where δ_i is the fraction of pollen reaching a focal stigma, γ_i the fraction that germinates or enters the pollen-tube pool, and τ_i the temporal overlap among competing grains.

Why this step

We split ψ_i into three multiplied fractions because each is a separate filter pollen must pass: arrive (δ), germinate/enter (γ), be present at the same time as rivals (τ). Independent filters in sequence multiply, exactly as passing three independent checkpoints multiplies their pass rates.

Substituting gives

$$K_i = \frac{\psi_i A_i}{O_i}. \quad (3)$$

Caution

A fixed deposition fraction such as $\psi = 0.02$ is useful for simulation but is not a biological constant. It is a calibration value.

4.2 Why the ratio matters

When $K_i \leq 1$, effective pollen is no larger than ovule number; there is little numerical pressure for grains to exclude one another. When $K_i > 1$, only a subset can fertilise, creating the opportunity for selection. One index of that opportunity is

$$\Omega_i = \mathbb{E} \left[\frac{(P_{if} - O_i)_+}{P_{if}} \right], \quad (x)_+ = \begin{cases} x, & x > 0, \\ 0, & x \leq 0. \end{cases} \quad (4)$$

The numerator counts grains that cannot win; the denominator turns it into a fraction.

Why this step

We use the positive part $(\cdot)_+$ because a pollen *deficit* does not create negative competition; when pollen is short the excess is zero, not negative. The expectation averages over flowers because even if the mean load does not exceed ovules, some flowers will, and those flowers still host competition.

Conclusion

K says whether competition is possible on average; Ω says how much pollen is excluded at the flower level. Both should be high in the strongest version of the model.

5 Sublinear pollen-load scaling

The model needs one scaling condition: effective pollen load should rise with ovule number more slowly than one-to-one. Write

$$P_{\text{eff},i} = \psi c O_i^\alpha, \quad (5)$$

with $c > 0$ and scaling exponent α .

Why this step

A power law O^α is the natural way to say “grows, but at a tunable rate”. One number decides the world: $\alpha = 1$ means pollen tracks ovules one-for-one, $\alpha < 1$ means pollen lags (sublinear), $\alpha > 1$ means pollen outpaces ovules.

Substitute into K_i and use the exponent rule:

$$K_i = \frac{\psi c O_i^\alpha}{O_i} = \psi c O_i^{\alpha-1}. \quad (6)$$

Rule used

Exponent rule for division: $\frac{x^a}{x^b} = x^{a-b}$, here with $a = \alpha$, $b = 1$.

Take logs of both sides:

$$\log K_i = \log(\psi c) + (\alpha - 1) \log O_i. \quad (7)$$

Rule used

Two log rules: product to sum, $\log(xy) = \log x + \log y$; power to front, $\log(x^p) = p \log x$.

Why this step

We take logs because a power law is a straight line on log–log axes. After logs, (7) reads “ $\log K$ is a line in $\log O$ with slope $\alpha - 1$ ”. The slope, not the intercept, carries the biology, turning “does K rise or fall?” into “is the slope positive or negative?”.

Differentiate; the constant intercept drops out:

$$\frac{\partial \log K_i}{\partial \log O_i} = \alpha - 1. \quad (8)$$

If $\alpha < 1$ then $\alpha - 1 < 0$, so the derivative is negative.

Conclusion

If effective pollen load scales sublinearly with ovule number, the pollen-per-ovule ratio falls as ovule number rises. Few-ovule lineages have the stronger competition regime. The single inequality $\alpha < 1$ is the hinge of the whole model.

5.1 The competition threshold

Set $K_i = 1$ in (6): $1 = \psi c O_i^{\alpha-1}$. Divide by ψc , use $\alpha - 1 = -(1 - \alpha)$ so $O^{\alpha-1} = 1/O^{1-\alpha}$, invert, then raise to $1/(1 - \alpha)$:

$$O^* = (\psi c)^{1/(1-\alpha)}. \quad (9)$$

Rule used

A negative exponent is a reciprocal, $x^{-p} = 1/x^p$. To undo a power x^q , raise to $1/q$, since $(x^q)^{1/q} = x$.

Why this step

We solve $K = 1$ because it is the exact tipping point: below it pollen is not limiting and competition is weak; above it pollen is in surplus and competition switches on. Solving for O^* turns the abstract condition into a concrete ovule number to compare against real species.

Conclusion

There is no universal ovule threshold. It depends on pollen production, transfer, germination, timing, and α .

6 Selection among pollen grains

6.1 Saved fraction

If $P > O$, only O grains win, so the winning fraction is $q = O/P$. Since $K = P/O$, i.e. $P = KO$,

$$q = \frac{O}{KO} = \frac{1}{K}. \quad (10)$$

Why this step

The saved fraction q is the chance a grain gets a seat. The clean result $q = 1/K$ says K is exactly the reciprocal of the winning chance. This is the bridge from a ratio to a selection problem, because “only the top q survive” is precisely truncation selection.

Conclusion

At $K = 2$ half the grains win; at $K = 10$ a tenth win. As K rises the winners come from a smaller, more extreme tail of the performance distribution.

6.2 Truncation selection

Let pollen performance $z \sim N(\bar{z}, \sigma_z^2)$. Standardise, $x = (z - \bar{z})/\sigma_z \sim N(0, 1)$.

Why this step

We standardise because the shape of selection cares only about how many standard deviations above the mean the winners sit, not the units of z . Working in standard-normal units lets us solve once and translate back by multiplying by σ_z .

The cutoff for the top fraction q is $t = \Phi^{-1}(1 - q)$.

Why this step

The winners are the top q , so the cutoff has q of the distribution above it, i.e. $1 - q$ below it; that is the quantile $\Phi^{-1}(1 - q)$. We invert Φ because we know the probability and want the cutoff that yields it.

The mean of the upper tail of a standard normal is the inverse Mills ratio, and $1 - \Phi(t) = q$:

$$\mathbb{E}[x \mid x > t] = \frac{\phi(t)}{1 - \Phi(t)} = \frac{\phi(t)}{q}. \quad (11)$$

Rule used

For a standard normal, the upper-tail mean above t is the inverse Mills ratio $\phi(t)/[1 - \Phi(t)]$; the denominator is the tail probability, here q .

Back on the original scale, $z = \bar{z} + \sigma_z x$, the selection differential is the winners' mean minus the original mean, and with $1/q = K$, $t = \Phi^{-1}(1 - 1/K)$:

$$S_z(K) = \begin{cases} 0, & K \leq 1, \\ \sigma_z K \phi \left[\Phi^{-1} \left(1 - \frac{1}{K} \right) \right], & K > 1. \end{cases} \quad (12)$$

Why this step

The differential is “how much better the winners are than average”; we subtract $\bar{z}$ because selection is the shift, not the level. As K rises, q falls, the cutoff moves right, and the surviving tail becomes more extreme, so S_z grows. This is “more competition, stronger pollen selection”.

Conclusion

Validated (Layer 1). Equation (12) matches a direct Monte Carlo and Falconer’s truncation-selection table to 3–4 decimals across $K = 1.5$ to 100. The selection engine is trustworthy before any RI is added.

6.3 From selection differential to allele-frequency change

The selection gradient is $\beta_z(K) = S_z(K)/\sigma_z^2$.

Rule used

A selection gradient is a differential divided by the variance, $\beta = S/\sigma^2$ (the Lande–Arnold relationship).

Why this step

Dividing by the variance converts “how different the winners are” into “how strongly fitness changes per unit of trait”, removing the effect of how spread out the trait already is.

For allele ℓ with pollen-performance effect a_ℓ , a male-function coefficient $s_\ell^{(m)} \approx a_\ell \beta_z(K)$, partly expressed at the whole-plant level as $s_\ell^{\text{eff}} = \kappa a_\ell \beta_z(K)$ with $0 < \kappa \leq 1$. The weak-selection recursion, with $K = \psi c O^{\alpha-1}$, is

$$\Delta p_\ell \approx p_\ell(1 - p_\ell) \kappa a_\ell \beta_z(\psi c O^{\alpha-1}). \quad (13)$$

Why this step

The factor $p_\ell(1 - p_\ell)$ appears because selection acts only on variation: an allele at frequency 0 or 1 cannot change, and change is fastest at intermediate frequency. Substituting $K(O)$ threads ovule number through to the speed of evolution.

Conclusion

If $\alpha < 1$, lower ovule number raises K , raising the differential, speeding allele-frequency change at pollen loci. Few-ovule lineages should evolve pollen traits faster, all else equal.

7 Maternal filtering

Selection among pollen changes male function. Directional RI also needs a maternal component. Let $F_i \geq 0$ be filter strictness. A strict filter improves offspring quality by rejecting unsuitable pollen, but reduces seed set when pollen is scarce. The balance depends on K .

7.1 A simple fitness model for the filter

Let the accepted fraction of pollen be e^{-F} , so accepted pollen per ovule is Ke^{-F} . Use a saturating seed-set function

$$S(K, F) = 1 - e^{-Ke^{-F}}. \quad (14)$$

Why this step

$1 - e^{-(\text{accepted pollen per ovule})}$ is the chance that at least one acceptable grain reaches an ovule under random (Poisson) arrival. It is near zero when accepted pollen is scarce and saturates at one when it is plentiful, exactly how seed set behaves. The filter enters as e^{-F} because each unit of strictness multiplies the accepted fraction down.

Quality benefit and direct cost are

$$Q(F) = 1 + b(1 - e^{-\lambda F}), \quad C(F) = e^{-c_F F^2/2}, \quad (15)$$

with $b, \lambda, c_F > 0$. Maternal fitness is the product

$$W_f(F, K) = S(K, F) Q(F) C(F). \quad (16)$$

Why this step

Fitness is a *product* because these are independent multiplicative requirements: seeds must set, *and* be good, *and* the machinery not cost too much. If any factor hits zero, fitness is zero, the signature of a product.

Take logs to turn the product into a sum:

$$\log W_f = \log S + \log Q - \frac{1}{2}c_F F^2. \quad (17)$$

Rule used

$\log(xyz) = \log x + \log y + \log z$. Maximising W and $\log W$ coincide when $W > 0$, because log is strictly increasing and preserves the maximiser.

Why this step

We take logs so the derivative splits into a clean sum of three derivatives, one per biological force: seed-set cost, quality benefit, maintenance cost. Sums differentiate far more easily than products.

7.2 The first-order condition

Let $x = Ke^{-F}$, so $\partial x/\partial F = -x$. By the chain rule $\partial S/\partial F = -xe^{-x}$, hence $\partial \log S/\partial F = -xe^{-x}/(1 - e^{-x})$. For Q , $\partial \log Q/\partial F = b\lambda e^{-\lambda F}/[1 + b(1 - e^{-\lambda F})]$, and the cost gives $-c_F F$. The optimum solves

$$-\frac{xe^{-x}}{1 - e^{-x}} + \frac{b\lambda e^{-\lambda F}}{1 + b(1 - e^{-\lambda F})} - c_F F = 0, \quad x = Ke^{-F}. \quad (18)$$

First term: seed-set cost (negative). Second: quality benefit (positive). Third: direct cost (negative).

Rule used

Chain rule $\frac{d}{dF} f(x(F)) = f'(x)x'(F)$ for S, Q ; log-derivative $\frac{d}{dF} \log g = g'/g$ converts each factor to its contribution to $\log W$.

Why this step

Setting the derivative to zero finds the strictness where the marginal benefit of one notch more choosiness exactly balances the marginal cost. Below it choosiness pays; above it choosiness costs more than it returns.

7.3 Why the optimal filter rises with competition

We prove $dF^*/dK > 0$ rather than assert it. Only the seed-set term depends on K ; write it as $g(x) = xe^{-x}/(1 - e^{-x}) = x/(e^x - 1)$, $x = Ke^{-F}$.

Rule used

Implicit-function rule: if the optimum solves $G(F^*, K) = 0$, then $dF^*/dK = -(\partial G/\partial K)/(\partial G/\partial F)$. At a maximum $\partial G/\partial F < 0$, so $\text{sign}(dF^*/dK) = \text{sign}(\partial G/\partial K)$.

With $G = \partial \log W_f/\partial F$, the only K -term is $-g(x)$, so $\partial G/\partial K = -g'(x)e^{-F}$. Since $g(x) = x/(e^x - 1)$ is strictly decreasing for $x > 0$, $g'(x) < 0$, hence $\partial G/\partial K > 0$ and

$$\frac{dF^*(K)}{dK} > 0. \quad (19)$$

Why this step

The biology: $g(x)$ is the marginal seed-set cost of rejecting a grain. In surplus (large x) a rejection barely dents seed set, so g is small; in scarcity each rejection risks an empty ovule, so g is large. Higher K means more surplus, cheaper choosiness, a higher optimal filter. Choosiness is a luxury surplus pollen pays for.

Conclusion

High K does not make a strict pistil inevitable; it makes one cheaper. If filtering improves quality or avoids poor pollen, selection then favours higher F . The relationship $dF^*/dK > 0$ is the maternal half of the mechanism.

8 Mating system as a modifier of competition

A selfing plant can pile pollen on its own stigma, giving a high *total* ratio. But if that pollen is one genotype, the opportunity for sexual selection among donors is low. We separate total from genetically diverse competition.

8.1 Partitioning the pollen pool

Write $P_{\text{eff},i} = P_{s,i} + P_{x,i} + P_{h,i}$ (self, outcross, heterospecific), with ratios $K_{\text{tot}}, K_{\text{out}}, K_{\text{bad}}$ formed by dividing each part by O_i . Donor diversity is

$$D_i^{\text{pollen}} = 1 - \sum_g \pi_{ig}^2, \quad (20)$$

π_{ig} being donor g 's share. Sexually effective competition is $K_{\text{sex},i} = K_{\text{tot},i} D_i^{\text{pollen}}$.

Rule used

$1 - \sum_g \pi_{ig}^2$ is the Simpson (Gini–Simpson) index: the chance two random grains come from different donors. It is 0 when one donor supplies all, near 1 when many contribute equally.

Why this step

Sexual selection needs genetically *different* competitors, not just many grains. Squaring shares and subtracting from one captures exactly that: a dominant genotype ($\pi \rightarrow 1$) gives sum-of-squares near one and diversity near zero. Self pollen inflates the count but not the diversity.

Conclusion

Use K_{sex} , not only K_{tot} , for sexual selection among distinct donors. Total pollen can be high in selfers while sexually effective competition stays low.

8.2 A maternal-filter model with self and outcross pollen

$$W_i(F) = \underbrace{\left[1 - \exp\left(-\frac{P_{x,i} + e^{-F_s} P_{s,i} + e^{-F_h} P_{h,i}}{O_i}\right) \right]}_{\text{seed-set assurance}} \underbrace{\left[1 - \iota_i \frac{e^{-F_s} P_{s,i}}{P_{x,i} + e^{-F_s} P_{s,i} + e^{-F_h} P_{h,i}} \right]}_{\text{quality protection}} \underbrace{e^{-c_F F^2/2}}_{\text{cost}}. \quad (21)$$

Here F_s filters self pollen, F_h heterospecific pollen, and ι_i is inbreeding depression.

Conclusion

Strict maternal filtering is most likely when outcross pollen is abundant, costly pollen is also present, and accepting costly pollen is expensive.

8.3 Regression forms with mating system

The directed success model can carry a competition-by-outcrossing interaction (key coefficient β_{Kt}):

$$\text{logit}(p_{i \leftarrow j}^{\text{pre}}) = \beta_0 + \beta_K \log K_i + \beta_t t_i + \beta_{Kt} (\log K_i) t_i + \beta_G G_{ij} + \dots \quad (22)$$

For asymmetry, using labels consistent with (41),

$$A_{ij} = \beta_0 + \beta_1 G_{ij} \Delta \log O_{ij} + \beta_4 \bar{t}_{ij} G_{ij} \Delta \log O_{ij} + \epsilon_{ij}, \quad \bar{t}_{ij} = \frac{1}{2}(t_i + t_j), \quad (23)$$

with $\Delta \log O_{ij} = \log O_j - \log O_i$ and prediction $\beta_4 > 0$.

Caution

Do not pool selfers, mixed-maters, and outcrossers without modelling mating system. A weak pooled signal could mean the mechanism is absent, or present only where genetically diverse pollen competition exists.

9 Prezygotic reproductive isolation

9.1 The reproductive isolation index

Use the Sobel–Chen index (Sobel and Chen, 2014), with $h_{i \leftarrow j} = H_{i \leftarrow j} / C_{ii}$:

$$R_{i \leftarrow j} = 1 - \frac{2H_{i \leftarrow j}}{H_{i \leftarrow j} + C_{ii}}. \quad (24)$$

Why this step

We divide H by the mother’s *own* conspecific success C_{ii} so a naturally low-fecundity mother is not mistaken for an isolated one. The index asks “how much worse is the heterospecific cross than this mother’s normal performance?”.

Substituting $H = hC_{ii}$ and simplifying (the C_{ii} cancels):

$$R_{i \leftarrow j} = \frac{1 - h_{i \leftarrow j}}{1 + h_{i \leftarrow j}}. \quad (25)$$

Why this step

The cancellation of C_{ii} top and bottom is the whole point: the index depends only on the ratio h , not on absolute fecundity, which makes it comparable across species with very different seed numbers.

9.2 Mechanistic content of relative success

Let D_{ij} be pollen-pistil mismatch, symmetric with $D_{ii} = 0$, and $F_i = F^*(K_i)$ a maternal property. Write

$$h_{i \leftarrow j} = \exp(-F_i D_{ij}). \quad (26)$$

Why this step

Relative success falls exponentially in $F_i D_{ij}$ because mismatch and strictness act like independent rejection hurdles: doubling either multiplies survival odds down, and repeated multiplicative survival is exponential. At $D_{ij} = 0$ (conspecific) the exponent is zero and $h = 1$, as it must be.

Substituting into (25) and using the half-angle identity,

$$R_{i \leftarrow j} = \frac{1 - e^{-F_i D_{ij}}}{1 + e^{-F_i D_{ij}}} = \tanh\left(\frac{F_i D_{ij}}{2}\right). \quad (27)$$

Rule used

Half-angle identity $\tanh(u/2) = \frac{1 - e^{-u}}{1 + e^{-u}}$. Proof: write $\tanh(u/2) = \frac{e^{u/2} - e^{-u/2}}{e^{u/2} + e^{-u/2}}$ and multiply top and bottom by $e^{-u/2}$. Here $u = F_i D_{ij}$.

Caution

Primary and sensitivity indices. The Sobel–Chen form (27) is the primary index. A simpler sensitivity index, $1 - H/C = 1 - e^{-F_i D_{ij}}$ (written $1 - h$), is reported alongside. They share the sign theorem and hump-shape but differ in scale (see (30)).

10 Directional asymmetry

10.1 Exact expression and the sign theorem

With $A_{ij} = R_{i \leftarrow j} - R_{j \leftarrow i}$,

$$A_{ij} = \tanh\left(\frac{F_i D_{ij}}{2}\right) - \tanh\left(\frac{F_j D_{ij}}{2}\right). \quad (28)$$

Because $\tanh$ is strictly increasing, for $D_{ij} > 0$,

$$\boxed{\text{sign}(A_{ij}) = \text{sign}(F_i - F_j)}. \quad (29)$$

Rule used

A strictly increasing function preserves order: $a > b \Rightarrow f(a) > f(b)$. Subtracting two ordered values keeps the sign of the original difference.

Why this step

This is the heart of the model. The pair shares one mismatch D_{ij} , so the only thing that can differ between cross directions is the maternal filter. Mismatch builds the wall; the maternal filter decides which side closes first. Symmetric divergence cannot pick a direction, a directional maternal trait can.

Conclusion

Validated (Layer 2). Over 200,000 random draws the simulated $\text{sign}(A_{ij})$ equalled $\text{sign}(F_i - F_j)$ with zero violations, for both indices.

10.2 Small-mismatch approximation

For small u , $\tanh(u) \approx u$, so $R_{i \leftarrow j} \approx F_i D_{ij}/2$ and

$$\boxed{A_{ij} \approx \frac{1}{2} D_{ij} (F_i - F_j)}. \quad (30)$$

Rule used

First-order Taylor at zero: $\tanh(u) = u - \frac{u^3}{3} + \dots \approx u$ for small u . Higher powers of a small number are negligible.

Why this step

At small divergence both directions are barely isolated, so the response is nearly linear in F and asymmetry is the filter difference scaled by the shared mismatch. The factor $\frac{1}{2}$ comes from the Sobel–Chen half-angle. The $1 - H/C$ index linearises to $A \approx D_{ij}(F_i - F_j)$ instead, i.e. twice the slope; the two agree in sign and shape and differ only by this constant.

Conclusion

Validated (Layer 2). Small-mismatch slopes recovered as 0.500 (Sobel–Chen) and 1.0 ($1 - H/C$), confirming the factor- $\frac{1}{2}$ distinction.

10.3 From filter strictness to ovule number

Chain $O_i \rightarrow K_i \rightarrow F_i$. We need $dF/d\log O = \frac{dF}{dK} \frac{dK}{d\log O}$.

Rule used

Chain rule $\frac{dF}{d\log O} = \frac{dF}{dK} \frac{dK}{d\log O}$, used because F depends on O only through K .

From (7), $d\log K/d\log O = \alpha - 1$. Converting proportional to absolute change with $dK = K d\log K$,

$$\frac{dK}{d\log O} = K(\alpha - 1), \quad \frac{dF}{d\log O} = F'(K) K(\alpha - 1). \quad (31)$$

Rule used

Logarithmic differentiation: $d\log K = dK/K$, so $\frac{dK}{dx} = K \frac{d\log K}{dx}$; it moves between proportional ($d\log K$) and absolute (dK) change.

Since $F'(K) > 0$ and $\alpha < 1$, this is negative. Define $\beta_F = -dF/d\log O = F'(K)K(1 - \alpha) > 0$, so

$$F_i - F_j \approx \beta_F (\log O_j - \log O_i). \quad (32)$$

Why this step

The negative sign says “more ovules, looser filter”, which under $\alpha < 1$ is the mechanism. We flip it and call it $\beta_F > 0$ for convenience, so the predictor $\log O_j - \log O_i$ enters with a positive coefficient when the few-ovule mother i has the stricter filter.

With $D_{ij} \approx \delta_D G_{ij}$, substitute into (30):

$$A_{ij} \approx \beta_1 G_{ij}(\log O_j - \log O_i), \quad \beta_1 = \frac{1}{2}\beta_F \delta_D > 0. \quad (33)$$

Caution

Natural log versus $\log_{10}$. The derivation uses natural logs, because $d \log K = dK/K$ holds for the natural log. The data section uses $\log_{10}$ for readability. They differ only by $\ln 10 \approx 2.303$, which rescales β_1 but changes neither its sign nor its significance. Use one base per analysis and report it.

Conclusion

The model predicts an *interaction*: asymmetry needs divergence to build mismatch *and* an ovule contrast to tilt the filter. Remove either and asymmetry vanishes. **Validated (Layer 2)**: regressing simulated A_{ij} on its predictors recovered $G_{ij}(\log O_j - \log O_i)$ as the dominant positive term, and the bare ovule main effect vanished as divergence shrank, exactly as (33) requires.

10.4 Why asymmetry peaks at intermediate divergence

With $F_i > F_j$, $A(0) = 0$ and $\lim_{D \rightarrow \infty} A(D) = 1 - 1 = 0$; in between the stricter filter saturates first, so A rises, peaks, and falls. Differentiating,

$$A'(D) = \frac{F_i}{2} \operatorname{sech}^2\left(\frac{F_i D}{2}\right) - \frac{F_j}{2} \operatorname{sech}^2\left(\frac{F_j D}{2}\right), \quad (34)$$

with the peak at $F_i \operatorname{sech}^2(F_i D/2) = F_j \operatorname{sech}^2(F_j D/2)$.

Rule used

$\frac{d}{du} \tanh u = \operatorname{sech}^2 u$; with inner factor $F/2$ the chain rule gives $\frac{F}{2} \operatorname{sech}^2(FD/2)$.

Why this step

Both endpoints give zero for the same reason: with no wall both crosses pass, with a complete wall both fail. Direction is visible only in the middle, where the stricter pistil has shut its side but the looser one has not. This is why the asymmetry test is a mid-divergence signal, and why pairs too close or too distant carry little information.

Working metaphor

Divergence builds a wall. The maternal filter decides which side closes first. Tiny wall: both pass. Complete wall: both fail. Direction is most visible in the middle.

11 Unilateral incompatibility as an alternative mechanism

Directional cross failure can also arise from the SI–SC rule: crosses often fail when the self-incompatible species is the mother and succeed in reverse (Lewis and Crowe, 1958; Wang and Filatov, 2023), with S-locus links in tomato (Li and Chetelat, 2015). In the same coordinates, with maternal rejection ρ_i and paternal resistance π_j ,

$$h_{i \leftarrow j}^{\text{UI}} = \exp[-\gamma(\rho_i - \pi_j)_+], \quad (x)_+ = \max(x, 0). \quad (35)$$

Why this step

The positive part encodes the rule: rejection bites only when the pistil out-guns the pollen. If pollen overcomes the pistil ($\pi_j \geq \rho_i$) the bracket is zero and the cross succeeds. This rectification is what makes UI genuinely one-sided rather than smoothly graded.

For small effects, $R_{i \leftarrow j}^{\text{UI}} \approx \frac{1}{2}\gamma(\rho_i - \pi_j)_+$, and if rejection tracks SI status,

$$A_{ij}^{\text{UI}} \approx \frac{1}{2}\gamma(s_i - s_j). \quad (36)$$

Caution

Equation (36) is a heuristic: it drops the rectifier $(\cdot)_+$ and the paternal term π_j from (35). Because of the rectifier, real UI can be strongly one-sided rather than an odd function of $s_i - s_j$; if anything this *sharpens* the contrast with the smooth, divergence-scaled ovule term. Treat (36) as the expected direction, not an exact form.

Conclusion

Ovule-gated asymmetry follows ovule contrast *and* divergence. Unilateral incompatibility follows SI-status contrast. Different fingerprints.

11.1 A discriminating regression

With $\Delta \log O_{ij} = \log O_j - \log O_i$ and $\Delta s_{ij} = s_i - s_j$,

$$A_{ij} = \beta_0 + \beta_1 [G_{ij} \Delta \log O_{ij}] + \beta_2 \Delta s_{ij} + \beta_3 \Delta \log O_{ij} + \epsilon_{ij}. \quad (37)$$

Labels are fixed hereafter: β_1 the ovule-gated interaction, β_2 the UI term, β_3 the bare ovule main effect. The ovule-gated fingerprint is $\beta_1 > 0$; the UI fingerprint is $\beta_2 > 0$.

Caution

Identifiability gate. If all few-ovule species are SI and all many-ovule species are SC, then $G_{ij}\Delta \log O_{ij}$ and Δs_{ij} are collinear and the mechanisms cannot be separated. Before fitting, measure their correlation; if strong, the analysis is still useful but cannot assign causality cleanly. This is the first gate to pass on the real data.

12 Postzygotic isolation as the contrast class

Model postzygotic isolation separately. Let $B_{ij} \sim \text{Poisson}(\lambda_B T_{ij}^\gamma)$ be the Dobzhansky–Muller load (T_{ij} divergence time; $\gamma = 1$ linear, $\gamma > 1$ accelerating), and

$$R_{ij}^{\text{post}} = 1 - e^{-hB_{ij}}, \quad R_{ij}^{\text{post}} \approx R_{ji}^{\text{post}}. \quad (38)$$

Why this step

Postzygotic isolation is symmetric because a hybrid carries both parents' alleles regardless of which was the mother; the incompatibilities that break it do not, to first order, care about cross direction. Exceptions (cytoplasmic, endosperm, parent-of-origin) exist but should not be in the baseline contrast.

Conclusion

The clean prediction is decoupling: prezygotic RI tracks maternal competition state and shows direction; postzygotic RI tracks divergence more strongly and shows weaker direction.

13 Empirical models

13.1 Use raw counts when possible

A proportion loses the denominator. For seed counts use

$$Y_{i \leftarrow j}^{\text{pre}} \sim \text{BetaBinomial}(N_{i \leftarrow j}, p_{i \leftarrow j}, \theta), \quad (39)$$

N the ovules exposed, Y the successful seeds.

Why this step

The Beta-Binomial, not a plain Binomial, allows scatter beyond coin-flipping: real crosses vary for reasons we have not modelled, and θ absorbs that overdispersion so uncertainty is not understated.

A directed model is

$$\text{logit}(p_{i \leftarrow j}) = \beta_0 + \beta_K \log K_i + \beta_G G_{ij} + \beta_{KG} (\log K_i) G_{ij} + a_i^{(M)} + a_j^{(P)} + a_{ij}^{(\text{pair})}, \quad (40)$$

with maternal, paternal, and pair effects carrying phylogenetic covariance. For reciprocal asymmetry (labels as in (37)),

$$A_{ij} = \beta_0 + \beta_1 G_{ij} \Delta \log O_{ij} + \beta_2 \Delta s_{ij} + \beta_4 \bar{t}_{ij} G_{ij} \Delta \log O_{ij} + \epsilon_{ij}, \quad (41)$$

with expected $\beta_1 > 0$ and $\beta_4 > 0$.

13.2 Phylogenetic structure

Species-pair data are not independent: a species recurs across crosses; a clade contributes many few-ovule species. Compare a non-phylogenetic reference, a low-overlap PI model, and a pairwise phylogenetic model, for which **phylopairs** is designed (Anderson et al., 2025).

Caution

Phylogenetic correction can erase signal if the signal is a repeated trait transition. The goal is not to erase ovule history but to ask whether the association survives shared ancestry and uneven replication.

14 Current empirical data structure

Three questions decide how strongly we can test the theory: do we have reciprocal directions; do both species have ovule estimates; are there enough within-family comparisons? The reciprocal-cross data exceed the strict PI/K2P subset, so the asymmetric test is the main axis with the PI/K2P subset as a conservative check.

Conclusion

The best immediate test is not the most restrictive complete-case test. It is the reciprocal-cross asymmetry test, with family, species reuse, measure type, ovule confidence, and phylogeny handled explicitly.

Data object	Size	Role
Table 1: cross outcomes	578 rows; 560 retained	Pair table: names, duplicate/removal flags, PI name, reciprocal intra/interspecific values, hybrid proportions, RI, measure.
Table 2: lineage traits	408 species	One ovule estimate per species (311 numeric, 97 missing), confidence, life history, mating system.
Table 2b: ovule provenance	5496 rows	Species/genus/family ovule evidence; curation and filling.
Genetic distances	141 rows; 133 unique	K2P distance (118 nonmissing, 23 missing; 8 duplicate rows).
PI tree	134 tips; 133 unique	One duplicated <i>Corymbia</i> tip to resolve before covariance modelling.

Table 1: Current data objects and their role.

Caution

The Table 1 description appears to repeat the same maternal direction for both interspecific columns. The audit treats the names as authoritative (`.._sp1_sp2` is sp1 mother $\times$ sp2 father; `.._sp2_sp1` the reverse). Confirm against the source before final modelling.

15 Exact asymmetric test for the current reciprocal crosses

With the four outcomes $C_{1\leftarrow 1}, C_{2\leftarrow 2}, C_{1\leftarrow 2}, C_{2\leftarrow 1}$ (symbol before the arrow is the mother), the exact Sobel–Chen RI simplifies as in (25):

$$R_{1\leftarrow 2}^{SC} = \frac{C_{1\leftarrow 1} - C_{1\leftarrow 2}}{C_{1\leftarrow 1} + C_{1\leftarrow 2}}, \quad R_{2\leftarrow 1}^{SC} = \frac{C_{2\leftarrow 2} - C_{2\leftarrow 1}}{C_{2\leftarrow 2} + C_{2\leftarrow 1}}, \quad (42)$$

and $A_{12}^{SC} = R_{1\leftarrow 2}^{SC} - R_{2\leftarrow 1}^{SC}$.

Rule used

Three checks: heterospecific=conspecific gives $R^{SC} = 0$; heterospecific= 0 gives $R^{SC} = 1$; heterospecific>conspecific gives $R^{SC} < 0$. Negative RI is not an error; the heterospecific cross simply beat the maternal baseline for that measure.

Use a signed $\log_{10}$ contrast $\Delta \log O_{12} = \log_{10} O_2 - \log_{10} O_1$ (positive when species 1 has fewer ovules). The first asymmetric regression is

$$A_{12}^{SC} = \beta_0 + \beta_1 G_{12} \Delta \log O_{12} + \beta_2 \Delta s_{12} + \gamma_{m[12]} + \epsilon_{12}, \quad (43)$$

γ_m a measurement-type effect, prediction $\beta_1 > 0$.

figures/reciprocal_asymmetry_funnel.png

Figure 1: Attrition to the reciprocal-cross subsets. The main asymmetric dataset has 305 comparisons across 41 families; the strict PI/K2P version has 63 across 35.

Dataset condition	Comparisons	Families
Retained pair rows	560	50
Both reciprocal interspecific directions	396	46
Directional RI possible (reciprocals + both intraspecific controls)	383	45
Asymmetry model-ready (directional RI + both ovule numbers)	305	41
... with nonzero ovule difference	217	27
High-confidence ovules (confidence ≤ 2 both species)	213	31
Strict PI/K2P asymmetry subset	63	35
Strict PI/K2P + high-confidence ovules	42	27

Table 2: Data attrition for the reciprocal-cross asymmetric test.

figures/family_reciprocal_asymmetry_top20.png

Figure 2: Family-level reciprocal-cross structure for the 20 largest families.

Solanaceae and Fabaceae supply 153 of the 305 asymmetry-ready comparisons, so a pooled model could become a two-family model in disguise. About 19 families can estimate a within-family ovule slope under a lenient rule, eight under a five-pair rule.

Conclusion

Do not fit a fixed slope per family. Use partial pooling: a global ovule-asymmetry slope with family slopes varying around it.

15.1 A partial-pooling model

$$A_{ij}^{SC} = \alpha + a_{f[ij]} + (\beta + b_{f[ij]}) G_{ij} [\log_{10} O_j - \log_{10} O_i] + \gamma_{m[ij]} + u_{s_1[ij]} + u_{s_2[ij]} + \epsilon_{ij}, \quad (44)$$

with $a_f \sim N(0, \sigma_a^2)$, $b_f \sim N(0, \sigma_b^2)$.

Why this step

Partial pooling treats each family slope as drawn from a shared distribution. Data-rich families pull toward their own estimate; sparse families borrow strength from the global mean. This avoids both one slope forced on all, and a noisy free slope for a family with two points.

Caution

Species reuse matters (*Leucaena* in Fabaceae, *Hydrophyllum* in Boraginaceae). Crossed species effects, phylogenetic covariance, or pairwise methods are needed to avoid overconfident intervals.

16 What the current data can and cannot claim

Strengths: reciprocal interspecific columns; cross outcomes separated from lineage traits; ovule confidence categories; a PI tree for a conservative check; many families. **Weaknesses:** the strict complete-case subset is only 63 comparisons; Solanaceae and Fabaceae dominate; some families have reciprocals but no ovule contrast; measurement types lack a common denominator; raw denominators are mostly absent, so a count model is a future target.

Conclusion

The data can test whether reciprocal asymmetry aligns with signed ovule contrast. They cannot yet prove the full chain from field pollen load to competition to filtering; that needs pollen dose, pollen-tube, ovule-exposure, and seed-count denominators.

17 Suggested tools for the first analysis

Three linked analyses. (1) **Broad reciprocal asymmetry** on all 305 pairs: response A^{SC} , predictor signed log ovule contrast, family partial pooling, measure type, confidence, crossed species effects. (2) **High-confidence ovules:** repeat with confidence ≤ 2 . (3) **Strict PI/K2P:** the 63 comparisons with the theoretical interaction $G_{ij}\Delta \log O_{ij}$. In R: `tidyverse`, `ape`, `phylolm/nlme`, `brms/MCMCglmm`, `phylopairs`. In Python: `pandas` for the audit; `bambi`, `pymc`, `statsmodels` for first models.

Caution

The first model should be boring. If the sign of the ovule-asymmetry effect is unstable in boring models, a complex Bayesian model will not rescue it. Complexity should protect against false certainty, not create the signal.

18 Missing data

Treat pollen load as latent. Sensitivity-sweep $\psi \in \{0.001, 0.003, 0.01, 0.02, 0.05, 0.10\}$ and $\alpha \in \{0.25, 0.50, 0.75, 1.00, 1.25\}$, computing $K_i = \psi c O_i^{\alpha-1}$, and ask whether the qualitative

result survives. A latent-variable form is

$$\log K_i = \mu_K + (\alpha - 1) \log O_i + \eta_t \text{logit}(t_i) + u_i + \epsilon_i, \quad (45)$$

u_i phylogenetic, ϵ_i lineage noise.

Conclusion

Missing pollen-load data become a calibration problem. The model is credible if the prediction's sign depends on $\alpha < 1$, not on one tuned ψ .

If only one cross direction exists, the data test average RI, not the maternal-filter prediction. For mating system, code $t_i \in \{\text{selfing, mixed, outcrossing}\}$ and re-run under alternative numeric codings (e.g. 0.05, 0.50, 0.95).

19 Simulation protocol

19.1 Layer 1: ecological competition without RI

Draw $P_{if} \sim \text{NegBinomial}(\mu_i, \theta_P)$, $\mu_i = \psi c O_i^\alpha$; compute $K = P/O$, Ω , and S_z from (12).

Conclusion

Status: built and validated. $S_z(K)$ matches Monte Carlo and Falconer's table; the $\psi \times \alpha \times O$ sweep confirms K falls with O for $\alpha < 1$, flat at $\alpha = 1$, reversed at $\alpha > 1$.

19.2 Layer 2: evolving maternal filters

Use the saturating filter

$$F_i^* = F_{\max} \frac{K_i}{K_i + K_{50}}, \quad (46)$$

simpler than the fitness model but keeping $dF^*/dK > 0$. Mismatch $D_{ij} = \delta_D G_{ij} + D_{ij}^{\text{arch}}$.

Conclusion

Status: built and validated. The directional core reproduces the sign theorem (zero violations in 2×10^5 draws), the $\frac{1}{2}$ -versus-1 linearisation, the hump in divergence, and the interaction $A_{ij} \approx \beta_1 G_{ij} \Delta \log O_{ij}$, for both indices. Architectures are deferred; Layer 2 fixes $D_{ij} = \delta_D G_{ij}$.

19.3 Layers 3–6

Layer 3 (architectures): independent; linked with hitchhiking $H(r, s) = s/(s + r)$; pleiotropic with association $\rho_{a,d}$. Expected order pleiotropy $>$ linkage $>$ independence.

Layer 4 (phylogeny): trees $n \in \{30, 60, 120, 240\}$; ovule number by Brownian motion, Ornstein–Uhlenbeck, and a discrete few/many transition model (the last counts independent transitions). **Layer 5 (crossing data):** $R = \tanh(FD/2)$, reciprocals, A_{ij} ; seeds $Y \sim \text{BetaBinomial}(N, 1 - R, \theta)$; postzygotic RI from (38). **Layer 6 (data damage):**

from full reciprocals with counts down to family-level states; for each, fit (41) or (37) and record bias, power, false-positive rate, sign recovery.

Conclusion

The simulation is judged by recovery, not appearance. The main target is a positive coefficient for $G_{ij}(\log O_j - \log O_i)$ after accounting for SI status and mating system.

20 Summary of the model

The model is one causal chain:

$$O_i \rightarrow K_i = \frac{P_{\text{eff},i}}{O_i} \rightarrow S_z(K_i) \rightarrow F_i^*(K_i) \rightarrow R_{i \leftarrow j}. \quad (47)$$

Under $P_{\text{eff},i} = \psi c O_i^\alpha$ with $\alpha < 1$, $K_i = \psi c O_i^{\alpha-1}$, so few-ovule lineages have higher K , stronger competition, and cheaper filtering. With mismatch between lineages, stricter maternal filtering yields directional prezygotic RI, summarised by

$$A_{ij} \approx \beta_1 G_{ij}(\log O_j - \log O_i), \quad \beta_1 > 0, \quad (48)$$

strongest in outcrossers and mixed-maters, weak in predominant selfers.

Conclusion

The claim is narrow enough to test. RI is not an inevitable clock; prezygotic RI can be regenerated when reproductive ecology places a lineage in a high pollen-competition state.

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